Joint-Embedding Predictive Architectures and Fluid-Dynamic Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize point-wise error in raw observation space, where unforecastable jitter and scale disparities produce heavy-tailed residuals and frequent false alarms under automated calibration. Joint-Embedding Predictive Architectures (JEPAs) avoid this by forecasting compact latent summaries of future windows, yet they still sum many prediction errors into one anomaly score, so one or two coordinates that drift between training and calibration can dominate the threshold and suppress point-wise F1. We pair a multi-horizon JEPA with a 224-bit circuit-breaker panel that replaces the scalar score and its tail-fitted threshold. Each of 32 whitened latent coordinates is expanded across four nested horizons into seven statistics (four horizon errors and their mean, spread, and endpoint difference), and each statistic receives its own breaker calibrated on healthy data with a drift guard. The breakers are statistically dependent, so alarm quorums are set from joint nominal trip counts. On 118 multivariate telemetry streams never used to tune the panel, a lighter three-horizon variant raises strictly causal point-wise F1 from 0.072 to 0.083 for scalar JEPAs to 0.143 and causal event recall from 51 to 73 percent, at the cost of a higher held-out false-alarm rate (8.3 versus 5.9 percent). The panel needs roughly a thousand calibration steps to hold its false-alarm rate, and it misses slow process ramps and most single-channel sensor freezes.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Fluid-dynamic decomposition, Boolean circuit-breaker panels, Representation learning.**

I. Introduction

Spacecraft, chemical plants, server farms, and wearable medical devices all report their health through dozens of sensor channels that never stop streaming [1]–[4]. Somewhere in that stream a fault begins: a valve sticks, a reaction wheel drifts, a patient's gait freezes. An automated monitor has to notice before people do, and it has to learn what healthy behavior looks like from normal operating history alone, because real faults are rare and seldom labeled in advance. It also has to stay quiet the rest of the time. A monitor that cries wolf every few minutes is soon ignored, and an ignored alarm protects no one.

Manuscript received October 15, 2026. This work was supported by the Department of Electrical Engineering and Computer Science at Florida Atlantic University. No prior conference presentation or thesis material is included in this submission. *(Corresponding author: Andre Williams.)*

Andre Williams, Imadeldin Mahgoub, and Hamid Akbarian are with the Department of Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL 33431 USA (e-mail: andrewilliam2018@fau.edu; mahgoubi@fau.edu; hakbaria@fau.edu).

Color versions of one or more figures in this article are available online at https://ieeexplore.ieee.org.

Why Today's Detectors Flood the Console: Leading deep detectors such as TimesNet [5], TranAD [6], and Anomaly Transformer [7] learn to reconstruct the raw sensor signal and raise an alarm wherever the reconstruction misses. Raw telemetry, however, is full of noise that no model can predict: sensor jitter, thermal drift, and channels recorded on very different scales. Summed across 25 to 55 channels, these small misses form an error signal with frequent extreme spikes (Table I). We calibrate each detector on healthy data with an automated tail rule, Streaming Peaks-Over-Threshold (SPOT) [8], and then run it on later healthy data, exactly as a deployed system would. The reconstructors raise false alarms during 22% to 31% of normal operation. At that rate the alarms carry almost no information: a random alerter firing just as often would catch 60.0% of real anomaly events, slightly more than the 58.8% that TimesNet catches.

The Hidden Weak Link: One Score, One Threshold: A more promising idea is to stop predicting the noise. Joint-Embedding Predictive Architectures (JEPAs) [9], [10] compress each window of telemetry into a short summary of 32 numbers and predict the summary of what comes next instead of every raw reading. Because the summary keeps the structure of the signal and discards most of the jitter, false alarms fall by an order of magnitude, to 0.6%–2.4% of healthy operation. Yet latent detectors keep one habit from their predecessors. At the final step they add all of their prediction errors into a single anomaly score and sound the alarm when that score crosses a threshold. Our analysis shows that this last step is where much of the signal is lost. If one or two of the 32 summary coordinates drift slightly between training and calibration, as routinely happens in long-running systems, they dominate the sum. We call these coordinates *hijackers*. The threshold must rise to tolerate them, and genuine faults on the remaining 30 coordinates no longer clear it. Real-time (causal) Point-F1, a standard measure that balances missed and false detections at every time step, falls to 0.050–0.052, and faults go unnoticed for more than 240 steps.

Replacing the Threshold with a Circuit-Breaker Panel: We remove that final step. Instead of one score and one threshold, our model, *Multi-Horizon Reynolds JEPA*, reports to a panel of 224 separately calibrated circuit breakers. The analogy is the breaker box in a building: a fault on one circuit trips its own switch. To build the panel we borrow an idea from fluid mechanics, by analogy only. Reynolds decomposition [11] splits a turbulent flow into a steady mean current and the fluctuations around it. We apply the same split to each of the

32 prediction errors as it evolves over four nested look-ahead horizons of 16, 32, 48, and 64 steps, which gives seven views of every coordinate: its error at each of the four horizons, its average drift, its fluctuation, and its acceleration at the newest edge of the window. Each of the resulting $7 \times 32 = 224$ dials has its own breaker, set from healthy calibration data. A guard computed only from nominal data damps dials that had already drifted before deployment, so a hijacker mostly trips its own breaker instead of raising one shared threshold for every coordinate. An alarm sounds when enough breakers trip together, or at once when one of 32 fast onset relays sees a coordinate's newest horizon and its acceleration dial trip simultaneously. No single anomaly score is ever formed, and no tail threshold is fitted to one. Upstream, the encoder subtracts a running average of recent summaries to cancel slow level shifts and is trained so that its predicted summaries stay decorrelated.

A. Contributions

We evaluate the approach on 141 telemetry streams from 12 public datasets spanning spacecraft, industrial plants, server clusters, and wearables. Our contributions are:

1) *A breaker panel in place of a thresholded score:* We replace the single anomaly score and its tail-fitted threshold with a 224-bit circuit-breaker panel built from Reynolds-decomposed latent errors, nominal-only hijack guards, and 32 fast onset relays (Section IV-D). Applied to the same trained network on 19 primary streams, the panel raises causal Point-F1 from 0.052 to 0.138 (17 wins, 0 losses, $p = 1.5{\times}10^{-5}$). On the 118 streams never used to tune the panel, a three-horizon variant raises causal Point-F1 from 0.072–0.083 to 0.143 (95 wins, 15 losses, $p = 1.6{\times}10^{-13}$) and causal event recall to 72.8%, at a higher held-out false-positive rate (FPR; 8.3% vs. 5.9%) (Tables II–V).
2) *Where the gains come from:* Matched-size ablations (272k–312k parameters) trace each step. A layer-normalization (LayerNorm) [12] bottleneck tames extreme error spikes, cutting excess kurtosis γ_2 from 6.10 to 0.91; predicting learned summaries instead of raw readings halves calibration drift; and barycenter centering cuts the share of false alarms caused by regime drift from 71.3% to 18.7%. A six-step ladder from one dial to 224, together with leave-one-out tests, isolates each dial family, the onset relays, prefix routing, and the hijack guards (Table II).
3) *Fewer false alarms and better recall at a fixed alarm budget:* On the 19 primary streams, latent prediction holds causal FPRs at 0.6%–1.7%, against 22.2%–24.3% for reconstructors. When every detector receives the same 1% false-alarm budget (set post hoc from test labels, as a diagnostic), the panel catches 58.4% of anomaly events versus 21.7% for TimesNet, and on five NASA in-range contextual sequences it reaches 0.658 buffered precision-recall area under the curve (PR-AUC) (Tables III and IV, Fig. 4).
4) *Limits, theory, and latency:* A sensor that freezes silently barely moves the latent summary, so the panel alone catches only 25.0%–29.2% of such events and a rolling variance monitor is needed to reach 44.4%–52.8% (Fig. 3); slow process ramps remain undetected. Proposition 1 bounds a spectral concentration ratio of the horizon fluctuations to $[\frac{1}{M-1}, 1]$, reweighting a scalar score by it gives no measurable benefit, the per-dial quantiles need about 10^3 calibration steps, and short horizons ($S \in \{1, 4\}$) remove the 63-step reporting lag for 1D latent scoring (Fig. 4(a)).

TABLE I
KEY TERMS.

Term	Meaning
Latent residual	Multi-horizon gap $\tilde{\mathbf{E}} \in \mathbb{R}^{4\times 32}$ between predicted and observed 32D window summaries.
224-bit breaker	Boolean panel of 7×32 per-coordinate tripwires and 32 onset relays avoiding 1D score thresholding.
Reynolds dials	Per-coordinate horizon mean $\lvert\bar{e}_c\rvert$ and fluctuation σ'_c across horizons (32 dials each).
Right-edge relay	Coincidence switch $r_c(t)$ requiring terminal sub-block $[48, 64]$ and endpoint change or fluctuation to trip together.
Hijack guard	Nominal-only per-dial multiplier $\gamma_j^{\text{cal}}\gamma_j^{\text{sc}}$ damping dials that drift on healthy calibration data.
Buffered / Causal	Retrospective prefix-window voting vs. real-time trailing attribution at $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later healthy residuals, measured by Kolmogorov–Smirnov distance D_{KS}.
Excess kurtosis	Tail heaviness γ_2 of residuals; large values indicate frequent extreme spikes.
Concentration ratio	Bounded ratio $R_{\text{LT}} \in [\frac{1}{M-1}, 1]$: share of horizon-fluctuation energy in the leading eigenmode.

Roadmap. Section II reviews related work, Section III details the 141-stream benchmark, Section IV formulates the architecture and 224-bit circuit-breaker panel, Section V defines the protocol, Sections VI–VII report ablations and results, and Section VIII concludes.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timestamps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [4] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [7] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that trigger severe false-alarm flooding under baseline shift [13].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space [14]. Contrastive detectors construct positive and negative pairs: temporal and contextual contrasting [15] and TS2Vec [16] maximize contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [10] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [17]

employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent prediction across 11 datasets, Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [18], [19] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} from a separate target encoder, preventing collapse via exponential moving average (EMA) target updates [20] or variance-covariance penalties such as VICReg [21], Barlow Twins [22], and batch whitening [23].

Fluid-Dynamic Decomposition and Multi-Perspective Circuit Breakers vs. Scalar Thresholding: Multi-sensor telemetry reflects physical dynamical trajectories [24]. Neural ordinary differential equations (ODEs) [25], Physics-Informed Neural Networks (PINNs) [26], and Hamiltonian or Lagrangian networks [27], [28] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [29]. Koopman and operator-theoretic models [30]–[34] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we borrow Reynolds decomposition [11] from turbulence modeling, by analogy, to split multi-horizon latent residuals into a per-coordinate mean, fluctuation, and endpoint acceleration. Existing detectors collapse high-dimensional residuals into a 1D scalar score and extrapolate a threshold via Pickands–Balkema–de Haan extreme-value theory [35] and Generalized Pareto Distribution (GPD) fitting in SPOT [8], where a single drifting coordinate under serial correlation [36] can hijack the scalar threshold. We replace 1D scalar collapse with a 224-bit multi-perspective Boolean circuit-breaker panel and evaluate all detectors using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm curves [13], [37], [38].

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [39], [40], TimeEval [13], and mTSBench [41], expanded to 141 qualifying multivariate streams across 12 datasets in our archive-wide evaluation. Thirty-three streams enter the primary multi-seed held-out protocol; 12 are reserved with explicit accounting below. The suite spans four physical domains: *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [42] (K=38), where NASA's ground-truth archive [4] classifies each SMAP/MSL anomaly sequence as either an out-of-range *point* fault or an in-range *contextual* fault (Table IV(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [39] sampled at 64 Hz (K=9, 72 gait-freezing episodes, where S=64 equals a 1 s horizon). *Industrial processes and smart infrastructure* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 testbed valve and heater faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry at 1-minute resolution (K=9, 37 hydrological anomalies), Metro I-94 hourly highway traffic (K=5), Room-Occupancy sensing [43] (K=5), CalIt2 building flow [40] (K=2), and six CICIDS network streams (K=72). Cohort 3 (Section VI-B) separately injects six controlled single-channel faults into six real streams.

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$ across the 33 primary streams) and labeled anomaly test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatlines, matching standardized archive subsets in prior benchmarks [4], [13].

Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by a 64-step isolation gap (Section V-A). Primary 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$) and $K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$ (Table III(c)); among the 12 reserved streams, five have $T_{\text{nom}} \in [764, 1{,}492]$ ($N_{\text{cal}} \leq 101$ across CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), six CICIDS streams have K=72, $N_{\text{test}} \geq 1.36 \times 10^5$, and GECCO ($K$=9, 1 held-out event) enters Cohorts 3–4. To eliminate subset-selection bias, Cohort 5 evaluates all 141 qualifying multivariate streams across 12 datasets ($N_{\text{cal}} \geq 100$ and ≥ 1 test anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 OPPORTUNITY, GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN) at three seeds (Table V); the 14 GHL streams (reported in Cohort 1) and six CICIDS streams also meet the rule but are excluded by dataset.

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [38]; and (ii) streams within SMAP, MSL, Daphnet, and GHL share hardware platforms (N=7 dataset clusters). We therefore report Dataset-Macro means, between-stream and within-stream seed standard deviations (SDs), zero-tuning leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table III(c)).

IV. Methodology: Joint-Embedding Predictive Architectures and 224-Bit Fluid-Dynamic Circuit-Breaker Panels

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to binary alarm decisions. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and partitions the stream at index t into a *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$) and a *suspect target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to decision timestamp $\tau = t + S - 1$). For $K > 1$, a causal depthwise 1D convolutional stem filters each channel prior to 1×1 mixing ($K \leq 1$ uses an identity bypass). Channels are decomposed across $J = 4$

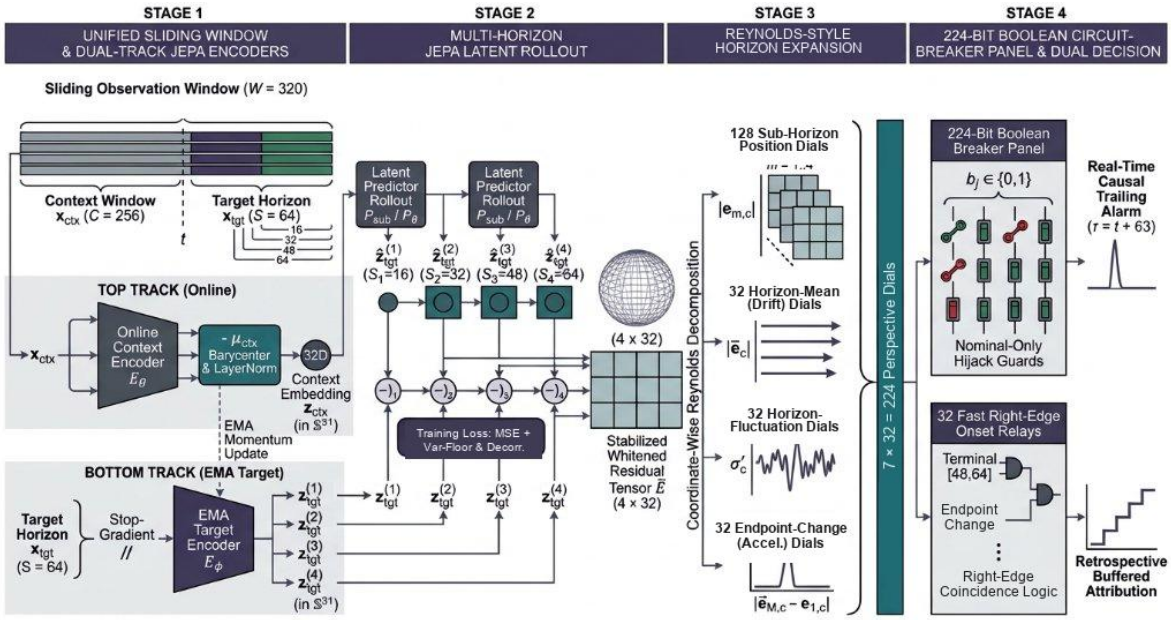


Fig. 1. Overview of the four-stage Multi-Horizon Reynolds JEPA and 224-bit circuit-breaker pipeline: (1) multiscale dyadic windowing, 3-depth TCN hierarchy tapping, and barycenter centering; (2) nested 4-horizon latent rollout under a variance-floor and decorrelation objective; (3) calibration-stabilized Mahalanobis whitening and Reynolds expansion into $7 \times 32 = 224$ dials; and (4) zero-collapse 224-bit Boolean circuit-breaker tripwires and 32 fast right-edge onset relays.

smooth dyadic shells on a 256-point real fast Fourier transform (rFFT) grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), processed by per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax router into a causal dilated temporal convolutional network (TCN) [44].

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit heavy-tailed sensor noise (Sections I–II), we map windows into a compact 32-dimensional latent space ($D = 32$) using a Joint-Embedding Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ onto the sphere $\sqrt{D}\mathbb{S}^{D-1}$, where target weights ϕ track online weights θ via an exponential moving average (EMA), $\phi_k \leftarrow m\phi_{k-1} + (1-m)\theta_k$ with fixed momentum $m = 0.996$ (0.995 in Cohorts 3–4) [18], [20]. Because deep dilated TCN blocks smooth out sharp onset spikes while non-stationary streams shift the latent mean, Multi-Horizon Reynolds JEPA introduces two structural operations inside both encoders:

1) *Multi-depth functional hierarchy tapping:* A tap head extracts $L_{\text{tap}} = 3$ intermediate feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B \times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 48$) from the shallow dyadic front-end, the mid-depth TCN block, and the terminal TCN block. For each tap ℓ, the channel-centered terminal state and temporal root-mean-square (RMS) energy are concatenated into $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B \times 6d_{\text{model}}}$, projected into $\mathbb{R}^D$, and fused with the deep representation $\mathbf{z}_{\text{deep}}$ via a bounded gate $g = 0.5\,\sigma(\theta_g) \le 0.5$:

$$\mathbf{z}_{\text{fused}} = (1-g)\,\mathbf{z}_{\text{deep}} + g\;\text{LN}\big(W_2\,\phi_{\text{GELU}}(\text{LN}(W_1\mathbf{h}_{\text{tap}}))\big). \tag{1}$$

2) *Synchronized barycenter centering:* An online running centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ updated at rate 0.05 on training batches is copied synchronously to the EMA target tap head and subtracted from both context and target fused vectors before terminal LayerNorm projection $z = \text{LN}(\mathbf{z}_{\text{fused}} - \bar{\boldsymbol{\mu}}_{\text{ctx}}) \in \sqrt{D}\mathbb{S}^{D-1}$, canceling common-mode baseline drift between windows.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Variance-Floor Decorrelation Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a single full-window forecast $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$ trained with predictive MSE plus a log-determinant volume term $-\frac{\beta}{D}\ln\det(\boldsymbol{\Sigma} + \epsilon\mathbf{I})$ on predicted and context batches ($\beta = 0.05$), a smooth relative of the VICReg variance-covariance penalties [21]. Terminal LayerNorm bounds representations on $\sqrt{D}\mathbb{S}^{D-1}$ and compresses residual kurtosis relative to observation-space reconstructors, but pooling the entire $S = 64$ target window into a single vector averages out short-lived sub-horizon transients.

To resolve sub-window dynamics without sacrificing the 64-step lookahead, Stage 2 of *Multi-Horizon Reynolds JEPA* partitions the target horizon $\mathbf{x}_{\text{tgt}}$ into $M = 4$ nested causal prefixes $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ of lengths $S_m = (m/M)S \in \{16, 32, 48, 64\}$. The EMA target encoder E_ϕ encodes each prefix into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. Starting from $\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$, a sub-horizon predictor P_{sub} rolls out sequential forecasts $\hat{z}_{\text{tgt}}^{(m)} = \text{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \dots, M-1\}$ using learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the full-horizon anchor P_θ forecasts $\hat{z}_{\text{tgt}}^{(M)} = \text{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ ($m < M$) and $w_M = \frac{1}{2}$, training minimizes multi-horizon mean-squared error (MSE) plus a variance-floor and decorrelation penalty:

$$\mathcal{L}_{\text{MHR}} = \sum_{m=1}^{M} \frac{w_m}{D}\Big(\big\|\hat{z}_{\text{tgt}}^{(m)} - \text{sg}\big(z_{\text{tgt}}^{(m)}\big)\big\|_2^2 + \big\|\boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big)\big\|_F^2\Big), \tag{2}$$

where $\mathbf{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ penalizes low variance and cross-covariance of centered batch predictions $\mathbf{Z}_c = \mathbf{Z} - \mathbf{1}_B \bar{\mathbf{z}}^\top$ with empirical covariance $\mathbf{\Sigma}(\mathbf{Z}) = (B-1)^{-1}\mathbf{Z}_c^\top \mathbf{Z}_c$ and marginal standard deviations $\sigma_d(\mathbf{Z}) = \sqrt{\Sigma_{dd}(\mathbf{Z}) + \epsilon}$ ($\gamma = 0.75$, $\epsilon = 10^{-4}$), defined by $[\mathbf{\Delta}_\gamma(\mathbf{Z})]_{dd} = \sqrt{\max(0, \gamma - \sigma_d(\mathbf{Z}))}$ and $[\mathbf{\Delta}_\gamma(\mathbf{Z})]_{dj} = \frac{1}{\sqrt{2}}\Sigma_{dj}(\mathbf{Z})$ ($d \neq j$). This is a variance floor plus decorrelation, not full whitening: it does not force unit variance, and the only whitening in the pipeline is the Ledoit–Wolf step of Stage 3. Unlike the log-determinant term of TS-JEPA, whose batch covariance is rank-deficient when $B < D$ (here $B = 16 < D = 32$) and stays finite only through ϵ, $\|\mathbf{\Delta}_\gamma(\mathbf{Z})\|_F^2$ remains informative for any $B \geq 2$ and raises effective representation rank $\mathrm{RankMe}(z_{\text{tgt}})$ from 11.6 to 19.8 [45].

Archive-wide variant: The 141-stream benchmark (Cohort 5) uses a lighter three-horizon model, *Speculative-JEPA*, with prefixes $\{16, 32, 64\}$ and the same rollout, trained with the TS-JEPA objective at each horizon and weights $(0.20, 0.25, 0.55)$ instead of (2). The panel below expects four horizons, so for this model the missing 48-step residual is filled with the mean of the 32- and 64-step residuals. Its 32 dials at $m = 3$ are then linear combinations of other dials, the panel carries 192 distinct measurements rather than 224, and the mean, fluctuation, and relay statistics include one interpolated horizon. All other cohorts use the four-horizon model.

C. Stage 3: Calibration-Stabilized Whitening and 224-Dial Reynolds-Style Expansion

Why does collapsing a multi-horizon residual tensor into a single 1D anomaly score degrade point-wise F1? Let $\mathbf{E}(t) = [\mathbf{e}_1(t), \ldots, \mathbf{e}_M(t)]^\top \in \mathbb{R}^{M\times D}$ stack the $M = 4$ raw sub-horizon prediction residuals $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ across all $MD = 128$ coordinates. Fitting the Ledoit–Wolf shrinkage precision matrix $\mathbf{\Omega} = (\hat{\mathbf{\Sigma}}_{\text{res}} + 10^{-3}\mathbf{I}_{MD})^{-1/2} \in \mathcal{S}_{++}^{MD}$ [46] and mean $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{MD}$ on nominal training windows produces Mahalanobis-whitened residuals $\mathbf{E}^{\text{white}}(t) \in \mathbb{R}^{M\times D}$ with unit training variance per coordinate. On unseen nominal calibration windows, however, one or two whitened directions often experience mild out-of-sample variance growth (*hijacker dials*) where $\mathbf{\Omega}$ amplifies small covariance shifts. When all 128 coordinates are summed or pooled into a 1D scalar score before calibration, those one or two hijacker dials dominate the upper calibration tail and push any scalar threshold to the extreme tip of the distribution, so that only isolated peak timestamps inside a true anomaly cross the threshold.

Stage 3 prevents both precision-matrix amplification and 1D coordinate collapse through two steps:

1) *Calibration-stabilized coordinate whitening:* On nominal calibration windows, we measure the whitened RMS $\mathrm{RMS}_{\text{cal}}^{\text{white}}(m,c)$ and raw RMS $\mathrm{RMS}_{\text{cal}}^{\text{raw}}(m,c)$ of each coordinate (m,c). Any coordinate whose whitened calibration RMS exceeds 1.10 is smoothly blended back toward its calibration-normalized raw residual $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/(\mathrm{RMS}_{\text{cal}}^{\text{raw}}(m,c) + 10^{-4})$ via the rational stability weight $w_{m,c}$:
$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t),$$
$$w_{m,c} = \frac{1}{1 + 2\max\big(0, \mathrm{RMS}_{\text{cal}}^{\text{white}}(m,c) - 1.10\big)^2}. \tag{3}$$
2) *Coordinate-wise Reynolds-style decomposition into $7D = 224$ dials:* In fluid turbulence, Reynolds decomposition [11] separates a flow field into a mean and zero-mean fluctuations. We borrow this split by analogy only: it is applied along the four-point lookahead axis $m \in \{1, \ldots, M\}$, not over time or a spatial field, and amounts to a per-coordinate mean, population standard deviation (denominator M), and endpoint difference across horizons. Rather than collapsing across the $D = 32$ latent coordinates, we expand the stabilized residual field $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into $7D = 224$ non-negative dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$:
   (a) *Sub-horizon flow-position magnitudes ($4D = 128$ dials):* $d_{\text{pos},m,c}(t) = |\tilde{e}_{m,c}(t)|$ for $m \in \{1,2,3,4\}$, isolating fault magnitude at prefix horizons $\{16, 32, 48, 64\}$.
   (b) *Horizon-mean (drift) dials ($D = 32$ dials):* $d_{\text{drift},c}(t) = |\bar{e}_c(t)| = \big|\frac{1}{M}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)\big|$, capturing persistent trajectory displacement across all four prefixes.
   (c) *Horizon-fluctuation dials ($D = 32$ dials):* $d_{\text{fluc},c}(t) = \sigma'_c(t) = \sqrt{\frac{1}{M}\sum_{m=1}^{M}(\tilde{e}_{m,c}(t) - \bar{e}_c(t))^2}$, capturing disagreement among horizons.
   (d) *Endpoint-change (acceleration) dials ($D = 32$ dials):* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$, measuring newly entering fault energy in the terminal sub-block $[48, 64]$ relative to the initial prefix $[0, 16]$.

Proposition 1 (Coordinate-Wise Energy Orthogonality, Acceleration Bound, and Spectral Concentration)**.** *Let $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) denote the stabilized multi-horizon residual field with horizon mean $\bar{\mathbf{e}} = M^{-1}\tilde{\mathbf{E}}^\top \mathbf{1}_M \in \mathbb{R}^D$ and zero-mean fluctuation matrix $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M \bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. Then: (i)* Coordinate-wise energy orthogonality and endpoint bound: *For every coordinate $c \in \{1, \ldots, D\}$, mean-square sub-horizon position energy splits orthogonally into horizon mean and fluctuation, $\frac{1}{M}\sum_{m=1}^{M} d_{pos,m,c}^2(t) = d_{drift,c}^2(t) + d_{fluc,c}^2(t)$, and the endpoint change satisfies $d_{accel,c}(t) \leq \sqrt{2M}\, d_{fluc,c}(t)$. (ii)* Spectral concentration bounds for the horizon Gram matrix: *Let $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ (an $M \times M$ matrix) have ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ (since $\mathbf{E}'^\top \mathbf{1}_M = \mathbf{0}$). For any $\gamma > -1/2$, the spectral concentration ratio, whose power-sum form follows the Lieb–Thirring moments [47],*
$$R_{LT}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^{M} \lambda_m^{\gamma+1/2}}, \tag{4}$$
is scale-invariant ($R_{LT}(c\mathbf{E}') = R_{LT}(\mathbf{E}')$ for $c > 0$) and satisfies for any $\mathbf{E}' \neq \mathbf{0}$:
$$\frac{1}{M-1} \leq R_{LT}(\mathbf{E}') \leq 1, \tag{5}$$
attaining $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) iff fluctuation energy is isotropic across $M-1$ modes, and 1 iff $\mathrm{rank}(\mathbf{E}') = 1$. These are elementary eigenvalue inequalities rather than a new Lieb–Thirring result and imply no physical wave propagation; the

implementation adds 10^{-12} *to the denominator only as a numerical guard near* $\mathbf{E}' = \mathbf{0}$.

Proof. For each coordinate c, expanding $\frac{1}{M}\sum_{m=1}^{M}(\bar{e}_c+e'_{m,c})^2$ with $\sum_{m=1}^{M} e'_{m,c} = 0$ yields the orthogonal energy identity; Cauchy–Schwarz on $\tilde{e}_{M,c} - \tilde{e}_{1,c} = e'_{M,c} - e'_{1,c}$ gives $d^2_{\text{accel},c} \le 2(e'^2_{M,c} + e'^2_{1,c}) \le 2M\, d^2_{\text{fluc},c}$, proving (i). For (ii), $\mathbf{G}\mathbf{1}_M = \mathbf{E}'(\mathbf{E}'^{\top}\mathbf{1}_M) = \mathbf{0}$ forces $\lambda_1 = 0$, leaving at most $M-1$ positive eigenvalues $\lambda_2 \le \cdots \le \lambda_M$ with $\lambda_M^{\gamma+1/2} \le \sum_{m=2}^{M}\lambda_m^{\gamma+1/2} \le (M-1)\lambda_M^{\gamma+1/2}$, which yields (5). The proposition records algebraic properties of the dials only; it does not show that the panel detects faults better than a scalar score, nor that reweighting by R_{LT} cannot help. Both are empirical questions (Section VI). □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

In Stage 4 (Fig. 1), each of the 224 dials receives its own binary circuit breaker calibrated strictly on nominal data. The breakers are separately calibrated but not statistically independent: all derive from the same 4×32 residual field, so position, drift, fluctuation, and acceleration dials of one coordinate are correlated by construction (Proposition 1(i)). On held-out nominal windows of the 19 primary streams, the dial correlation spectrum has a median effective rank of 65 of 224 (range 13–129), and simultaneous trip counts have 7.2× the variance of an independence model, so the quorums below are calibrated on joint nominal trip counts:

1) *Per-dial standardization and nominal hijack guards:* Each dial $j \in \{1,\dots,224\}$ (and $j \in \{1,\dots,128\}$ in the 128-bit sub-block panel) is standardized by its nominal calibration median m_j and scaled median absolute deviation (MAD) $s_j = 1.4826\ \text{MAD}(d_{j,\text{cal}}) + 10^{-4}$ into a non-negative excess score $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine any drifting dial individually without desensitizing the remaining dials, each tripwire τ_j scales its 98.5th calibration percentile by a two-factor nominal hijack guard $\gamma_j = \gamma_j^{\text{cal}}\gamma_j^{\text{sc}}$:
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\,\gamma_j^{\text{cal}}\,\gamma_j^{\text{sc}}, \quad (6)$$
where γ_j^{cal} equals the ratio of second-half to first-half calibration 98th percentiles whenever that ratio exceeds 1.35 (and 1 otherwise), and the cross-dial scale guard γ_j^{sc}, computed from calibration statistics only, equals $(m_j + s_j)/(1.5\, s_{\text{group},j})$ whenever that ratio exceeds 1.25, where $s_{\text{group},j}$ is the median of $m_j + s_j$ over the dial's perspective group (and 1 otherwise).
2) *32 fast right-edge onset relays:* To catch newly entering faults in the terminal sub-block $[48, 64]$ without waiting for earlier prefixes to saturate, each coordinate $c \in \{1,\dots,32\}$ defines a stricter edge switch $b^{\text{edge}}_{p,c}(t) = \mathbb{I}\big(z_{p,c}(t) > q_{0.993}(z_{p,c,\text{cal}})\,\gamma_{p,c}^{1.25}\big)$ across perspectives p and fires its onset relay $r_c(t) \in \{0,1\}$ when the terminal sub-block ($m = 4$) coincides with convective acceleration, turbulent fluctuation, or the third sub-block ($m = 3$):
$$r_c(t) = b^{\text{edge}}_{\text{pos},4,c}(t) \wedge \big(b^{\text{edge}}_{\text{accel},c}(t) \vee b^{\text{edge}}_{\text{fluc},c}(t) \vee b^{\text{edge}}_{\text{pos},3,c}(t)\big). \quad (7)$$
3) *Causal trailing vs. prefix-routed buffered attribution and Boolean quorums:* Under *strictly causal trailing attribution*, window switches at index t are emitted solely at the right endpoint $\tau = t + S - 1$, which observes only $\mathbf{X}[t - C : \tau, :]$ ($\mathcal{F}_\tau$-measurable, with structural emission lag $S - 1 = 63$ steps vs. empirical detection delay $\Delta t = t_{\text{alm}} - t^*$, Fig. 2). Under *retrospective buffered attribution*, each perspective group votes only over the exact time steps inside $[t, t+S)$ that it physically observes: $[t, t+S_m)$ for sub-block $m \in \{1,2,3,4\}$, $[t, t+S)$ for mean drift and fluctuation, and $[t+S_3, t+S_4)$ for right-edge acceleration. Summing active switches at timestamp τ yields counts $N_{224}(\tau) = \sum_{j=1}^{224}\bar{b}_j(\tau)$, $N_{128}(\tau) = \sum_{j=1}^{128}\bar{b}_{128,j}(\tau)$, and $N_{\text{edge}}(\tau) = \sum_{c=1}^{32}\bar{r}_c(\tau)$, with combined count $N_{\text{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\text{edge}}(\tau)$ used for threshold-free PR-AUC ranking. The 128-bit counter reuses the 128 position dials of the 224-bit panel with the same thresholds: under causal attribution it is a sub-count of N_{224} that up-weights the position dials, and it differs from that sub-count only under buffered attribution, where it votes uniformly over $[t, t+S)$ instead of by prefix. Binary alarms trigger when any counter exceeds its nominal calibration quorum:
$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \ \vee\ N_{128}(\tau) > K_{128} \ \vee\ N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \quad (8)$$
where $K_{\text{edge}} = \max\big(1.25\, q_{0.999}(N_{\text{edge,cal}}),\, 2.5\big)$, *224-Bit Breaker* sets $K_{224} = \max\big(1.12\, q_{0.995}(N_{224,\text{cal}}),\, 5.0\big)$ and $K_{128} = \max\big(1.12\, q_{0.995}(N_{128,\text{cal}}),\, 3.5\big)$, and *224-Bit Strict* sets $K_{224} = \max\big(1.18\, q_{0.9985}(N_{224,\text{cal}}),\, 5.5\big)$ and $K_{128} = \max\big(1.20\, q_{0.9985}(N_{128,\text{cal}}),\, 4.0\big)$.

1D Scalar Collapse Controls (for Ablation Comparison): To isolate the gains of the 224-bit panel against legacy 1D scalar collapse on the identical trained backbones (Section VI), we evaluate the coordinate saliency gate $g_d(\mathbf{v}) = (1-\alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}$ with softmax weights $w_i \propto \exp(|v_i|/(\tau_g\sigma_{\mathbf{v}}))$ ($\sigma_{\mathbf{v}} = \|\mathbf{v}\|_2/\sqrt{d} + 10^{-6}$, $\alpha = \tau_g = 0.5$) . Single-horizon TS-JEPA scores $\mathcal{S}_{\text{TS}}(t) = g_D(\mathbf{e}(t))$, whereas 1D Multi-Horizon Reynolds JEPA scores the multi-horizon envelope $\mathcal{S}_{\text{env}}(t) = \sqrt{\frac{1}{2}(g_{MD}(\tilde{\mathbf{e}}(t))^2 + \max_m g_D(\tilde{\mathbf{e}}_m(t))^2)}$ modulated by $\mathcal{S}_{\text{LT}}(t) = \mathcal{S}_{\text{env}}(t)\big(1+\alpha_{\text{LT}}\max(0, \tanh(0.5\, z_{\text{LT}}(t)))\big)$ ($\alpha_{\text{LT}} = 0.12$, or $\alpha_{\text{LT}} = 0$ for the unmodulated envelope), calibrated via scalar Generalized Pareto Distribution (GPD) tail fitting in SPOT [8], [35], [48] ($u = p_{98}, q = 10^{-3}$).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup, Two-Stage Nominal Calibration, and Statistical Design

All architectures operate on an identical streaming pipeline with context window $C = 256$ and suspect target horizon $S = 64$ ($W = C + S = 320$), trained on nominal windows at stride 16 and evaluated at stride 1. Context and target windows are encoded into $D = 32$ spherical representations by the multiscale frequency front-end and causal dilated TCN, fused with the three-depth functional hierarchy tap head (1), and centered by the online running context barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$ prior to terminal LayerNorm. Models are trained on nominal sequences using AdamW [49] (learning rate 10^{-3}, weight

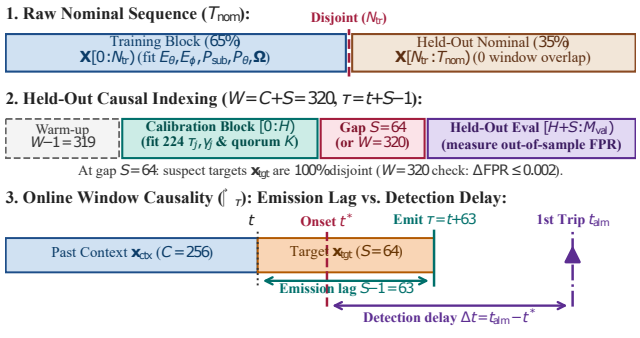


Fig. 2. Two-stage held-out evaluation protocol (C=256, S=64, W=320): (Row 1) chronological train/validation split, (Row 2) boundary-isolated nominal calibration and evaluation blocks fitting the 224 dial tripwires τ_j, hijack guards γ_j, and quorums ($K_{224}, K_{128}, K_{\text{edge}}$), and (Row 3) structural emission lag ($S-1$) versus empirical fault-detection delay (Δt).

decay 10^{-4}) with zero per-dataset hyperparameter tuning. Budgets are fixed within each comparison but differ by cohort: batch 16 at constant learning rate for 20 (Cohort 1) or 15 epochs (Cohort 2); batch 32 with gradient clipping at 1.0 for 10–12 epochs (Cohorts 3–4); and batch 16 for at most 12 epochs with cosine annealing to 10^{-5}, gradient clipping, and early stopping with patience 4 (Cohort 5). Cohort 2 reuses cached whitened residuals and therefore omits the blend in (3).

a) Two-Stage Nominal Calibration Protocol and Boundary Isolation: To evaluate autonomous alarm calibration without test-label leakage or train-to-calibration circularity, each stream's nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$):

1) *Stage A (Training fit on $\mathbf{X}_{tr}$):* Channel standardization parameters, encoder/predictor weights, online context barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and the multi-horizon shrinkage precision matrix $\mathbf{\Omega} \in \mathcal{S}_{++}^{MD}$ are fit strictly on $\mathbf{X}_{\text{tr}}$, ensuring zero raw-sample or window overlap across N_{tr}. In Cohort 5, early stopping selects the checkpoint by unsupervised predictive loss on the calibration block, so that block serves both checkpoint selection and threshold fitting; no labels are involved, all Cohort 5 arms share the rule, and the held-out block and test stream play no role in training. A validation partition disjoint from calibration would remove this reuse.
2) *Stage B (Circuit-breaker calibration and held-out verification on $\mathbf{X}_{val}$):* Sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2), yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid window timestamps after the $(W-1)$-step context warm-up. Splitting at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $[0 : H)$ and the *held-out evaluation block* $[H + S : M_{\text{val}})$, separated by an $S = 64$-step isolation gap $[H : H + S)$ so the last calibration target and first evaluation target share zero target observations. On $[0 : H)$, the panel fits its calibration-stabilized whitening weights $w_{m,c}$ (3), per-dial medians m_j and MAD scales s_j, two-factor hijack guards $\gamma_j = \gamma_j^{\text{cal}} \gamma_j^{\text{sc}}$, switch thresholds τ_j (6), and Boolean quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (8) (or scalar τ_{SPOT} for 1D baselines), which are then frozen and evaluated on $[H + S : M_{\text{val}})$ and on the test stream. A full-window isolation check ($G = W = 320$, excluding $[H : H + 320)$ across the seven streams with $M_{\text{val}}/2 > 420$) confirms that shared passive context changes causal held-out FPR by less than 0.25 percentage points.

b) Independent Statistical Units, Multi-Seed Replication, and Hypothesis Testing: Models are replicated across multiple random seeds ($\{42, 123, 456\}$ for all 224-bit circuit-breaker, ablation, and 141-stream benchmarks; $\{42, 123, 456, 7, 2024\}$ for 1D primary baselines). Seeds are averaged within each stream first (SD_{seed}) so that stream-level tests treat the telemetry stream ($n = 19$, $n = 33$, or $n = 141$) as the independent statistical unit, while domain-level tests treat the dataset family ($N = 7$ or $N = 12$) as the independent cluster:

1) *Paired Wilcoxon signed-rank tests and false-discovery-rate control:* We evaluate paired stream differences via the two-sided Wilcoxon signed-rank test without Gaussian assumptions, adjust p-values across families using the Benjamini–Hochberg false-discovery-rate (FDR) procedure [50] ($q_{\text{FDR}} = 0.05$), count $|\Delta| \leq 10^{-4}$ as ties, report 95% stream-resampled bootstrap confidence intervals (CIs; 2,000 resamples), and cross-check at the dataset-cluster level via exact sign-flip permutation tests and zero-tuning LODO evaluation.
2) *Ablations:* On the same trained backbones ($n = 19$ streams, 57 runs) we evaluate the expansion ladder ($1 \rightarrow 224$ dials) and leave-one-out removal of relays, prefix routing, hijack guards, or Boolean thresholds (Table II(a),(b)); null comparisons are reported as paired effect sizes with 95% bootstrap CIs, without equivalence claims.
3) *Tail and drift diagnostics:* On nominal windows we measure excess kurtosis $\gamma_2 = \mathbb{E}[(s-\mu)^4]/\sigma^4 - 3$ and the two-sample Kolmogorov–Smirnov (KS) statistic $D_{\text{KS}} = \sup_x |F_{\text{cal}}(x) - F_{\text{hold}}(x)|$ between calibration and held-out blocks.

B. Baseline Fairness Audits, Benchmark Cohorts, and Computational Footprint

Two audits check that reconstructor results are not artifacts of the shared $W = 320$ window: at their native windows on five streams, TranAD ($W = 10$) and TimesNet ($W = 96$) reach 0.164 and 0.218 Point-F1 but keep heavy tails ($\gamma_2 > 98$) and 28.4%–44.1% nominal FPR; and reconstructors replicate their published point-adjusted F1 on SMD and MSL. Table III(d) adds a 130-stream cross-check against NCAD, DCdetector, and Anomaly Transformer.

Experiments span five cohorts: *Cohort 1* (33 primary streams: 19 non-GHL and 14 GHL, Table III(a)–(c)); *Cohort 2* (19-stream perspective expansion ladder, circuit-breaker component ablation, 15-arm matched-backbone comparison, and operating-curve suite, Table II(a)–(c) and Figs. 3(a) and 4(b)); *Cohort 3* (6-stream single-channel quiet-fault injection and 224-bit hybrid variance-collapse monitor suite, Fig. 3(b)); *Cohort 4* (5-stream horizon-latency sweep across $S \in \{1, 4, 16, 32, 64\}$ and 3-seed 224-bit dyadic front-end ablation, Fig. 4(a) and Section VI-C); and *Cohort 5* (141-stream, 3-seed archive-wide benchmark across 12 multivariate datasets comparing 1D JEPAs against the panel on the three-horizon Speculative-JEPA with 192 distinct dials, Table V),

plus a separate single-seed screen of external baselines on 130 streams in 11 datasets (Table III(d)) that is never paired with Cohort 5 (91 shared streams; the other 39 are GHL, CICIDS, or fail $N_{\text{cal}} \geq 100$). The panel's calibration constants in (6)–(8) were chosen during development by inspecting test metrics on 23 streams (the 19 non-GHL primary streams plus SMAP T-2, MSL P-10, Daphnet S09R01E2, and OPPORTUNITY S1-ADL3), so Cohort 2 results are development-set results; the constants were then frozen, and the 118 Cohort 5 streams never used in development serve as the held-out test of the panel. At $K = 25$ channels, online parameter counts are matched (272k for TS-JEPA, 312k for Multi-Horizon Reynolds JEPA, 304k–307k for TCN observation forecasters, 649k for TimesNet, and 206k for TranAD; the 224-bit circuit-breaker panel adds zero trainable parameters). Runs executed on PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.88 ms/window, 188 MB memory); code, stream manifests, split settings, and analysis scripts are at https://github.com/quaid281/NCAD.

C. Multi-Tier Operational Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [13], [37], we evaluate detectors across three tiers: (i) *primary accuracy and false-alarm outcomes* comprising realized held-out nominal FPR (FPR_{hold}), unadjusted Point-F1, and threshold-free PR-AUC (computed from the total active switch count $N_{\text{tot}}(\tau)$ for the 224-bit panel); (ii) *operating characteristics* evaluating event recall across autonomous circuit-breaker operating points (*224-Bit Strict* and *224-Bit Breaker*), deployable calibration quantiles (q^*, Fig. 4(b)), and, as a diagnostic only, post-hoc matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table II(c)) whose thresholds use test labels and cannot be set before deployment; and (iii) *secondary operational endpoints* reporting unweighted Dataset-Macro Point-F1, event recall (ER, fraction of contiguous anomaly segments with ≥ 1 alarm), mean detection delay $\Delta t = t_{\text{alm}} - t^*$ over detected events, and the constant-alarm reference $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: Perspective Expansion Ladder and Circuit-Breaker Component Analysis

How do each of the seven fluid-dynamic perspectives and three calibration rules contribute to the 224-bit circuit-breaker panel on the identical trained backbone? Table II(a),(b) and Fig. 3(a) evaluate the step-by-step perspective expansion ladder and leave-one-out component ablations across all 19 non-GHL streams (57 runs):

1) *From 1D scalar collapse to 32-bit and 128-bit sub-horizon switches (Table II(a)):* Collapsing the 128 whitened coordinates into a single 1D score (with or without R_{LT} modulation; $\alpha_{\text{LT}} = 0.12$ changes causal F1 by -0.005, 95% CI $[-0.012, 0.000]$) allows one or two drifting coordinates to inflate the scalar threshold, restricting causal Point-F1 to 0.052 and delay to 246.6 steps. Replacing 1D collapse with 32 per-coordinate switches on the terminal horizon ($m = 4$) nearly doubles causal Point-F1 to 0.101, and expanding across all four nested sub-horizons (*128-Bit Sub-Horizons*) lifts causal Point-F1 to 0.129, buffered Point-F1 from 0.137 to 0.192, and buffered PR-AUC from 0.370 to 0.402. These two rungs are plain per-coordinate threshold-and-quorum detectors, so they are the simple multi-coordinate baselines against which the Reynolds statistics are measured.
2) *Adding horizon mean, fluctuation, and endpoint change:* The 32 drift dials (*160-Bit*) halve causal delay (146.2 $\rightarrow$ 69.1 steps); the 32 fluctuation dials (*192-Bit*) lift buffered Point-F1 to 0.203 but lengthen causal delay to 179.0 steps, so no rung improves every metric. Adding the 32 endpoint-change dials and onset relays (*224-Bit Breaker*) reaches 0.138 causal Point-F1 (17 wins / 0 losses over 1D collapse, $p = 1.5\times10^{-5}$), 0.228 buffered Point-F1 ($p = 0.026$), 0.411 buffered PR-AUC, 60.7% causal and 63.1% buffered ER, and 71.8-step buffered delay.
3) *Leave-one-out component ablations (Table II(b)):* Removing the 32 fast right-edge onset relays slows causal detection delay from 105.9 to 123.7 steps and lowers causal Point-F1 from 0.138 to 0.135. Replacing physical prefix-horizon routing with uniform 64-step window smearing degrades buffered ER from 63.1% to 58.4%, buffered Point-F1 from 0.228 to 0.223, and buffered delay from 71.8 to 88.6 steps. Removing per-dial hijack guards ($\gamma_j = 1$) lowers causal and buffered F1 to 0.137 and 0.225, whereas adding a soft attention pooling gate provides no gain over discrete Boolean switches (0.137 causal and 0.226 buffered F1).

B. Ablation 2: Single-Channel Quiet Faults and the 224-Bit Hybrid Variance-Collapse Monitor

When one sensor out of K freezes at a nominal level while the remaining $K-1$ channels fluctuate normally, the frozen channel perturbs the pooled pre-LayerNorm vector by only $O(1/K)$, and LayerNorm rescales the embedding onto $\sqrt{D}\mathbb{S}^{D-1}$ using the remaining active channels. Across six single-channel fault classes injected into six multivariate streams (Fig. 3(b)), 1D TS-JEPA detects only 19.4% of quiet-fault events (flatlines, stuck sensors, and dropouts) with $< 3\%$ point recall. Because the 224-bit panel thresholds all 32 latent coordinates separately, standalone *Multi-Horizon Reynolds 224-Bit Breaker* lifts quiet-fault event recall to 25.0%–29.2% (33.3% on $+5\sigma$ spikes) and increases point recall more than seven-fold to 14.3%–25.6%. A logical union with a parameter-free rolling variance-collapse monitor (how far each channel's 32-step trailing standard deviation drops below its nominal median) lifts event recall to 44.4%–52.8% and point recall to 24.3%–35.6%. The JEPA representation alone is insufficient for isolated sensor freezes, and this evidence rests on six injected streams.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

To isolate the effect of the 4-shell rFFT front-end under the 224-bit circuit-breaker panel, we train Multi-Horizon Reynolds JEPA with and without dyadic frequency-shell splitting across five benchmark streams (`SMAP P-3`, `MSL M-1`,

TABLE II
MULTI-PERSPECTIVE CIRCUIT-BREAKER EXPANSION LADDER, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY (n=19 STREAMS).

(a) Perspective Expansion Ladder (1 → 224 Dials, n=19)

Perspective Expansion Stage	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
1D Scalar Collapse (1 Dial)	.052	.137	.291	.370	.398	.377	246.6	402.7
32-Bit Full Horizon (m=4)	.101	.119	.266	.366	**.643**	.236	114.4	128.8
128-Bit Sub-Horizons (m=1..4)	.129	.192	.310	.402	.590	.525	146.2	514.5
160-Bit (+32 Reynolds Drift)	.132	.188	.311	.404	.585	.530	**69.1**	112.4
192-Bit (+32 Reynolds Fluc.)	.131	.203	**.312**	.408	.575	.567	179.0	217.3
224-Bit Breaker (+32 Accel./Relays)	**.138***	**.228**†	**.312**	**.411**	.607	**.631**	105.9	**71.8**

*Wilcoxon vs. 1D Scalar Collapse: 17/0 wins (p=1.5×10^{-5}); †buffered F1 p=0.026. Fixed-backbone 1D α_{LT}=0.12 vs. 0: causal F1 Δ=−0.005 (95% CI [−0.012, 0.000], p=0.13).

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

Component / Quorum Variant	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
224-Bit Breaker (Full Panel)	**.138**	**.228**	.312	.411	**.607**	**.631**	105.9	**71.8**
224-Bit Strict (High Quorum K_{strict})	.116	.190	.312	.411	.534	.545	248.0	97.3
224-Bit + Soft Attention	.137	.226	.311	.410	**.607**	.628	106.8	73.4
w/o 32 Right-Edge Relays	.135	**.228**	**.313**	**.413**	.602	**.631**	123.7	72.5
w/o Prefix Routing (Unif. Smear)	**.138**	.223	.312	.407	**.607**	.584	105.9	88.6
w/o Hijack Guards (γ_j=1)	.137	.225	.312	.411	.603	.623	**101.6**	75.6

Leave-one-out component ablations evaluated on the identical 57 cached Reynolds multi-horizon residual fields (n=19 streams × 3 seeds).

(c) Matched-Backbone Architectural Comparison (n=19 Streams)

Architecture Arm	Tail / Drift γ_2	Tail / Drift D_{KS}	Held-Out FPR Caus.	Held-Out FPR Buf.	Point-F1 Caus.	Point-F1 Buf.	PR-AUC Caus.	PR-AUC Buf.	Causal Event Recall Auto.	Causal Event Recall q^*=0.05	Causal Event Recall f^*=0.01	Causal Event Recall f^*=0.05
Reynolds 224-Bit Breaker	2.30	0.179	0.017	0.132	0.138	0.228	**0.312**	**0.411**	**0.607**	0.774	**0.584**	**0.728**
Reynolds 224-Bit Strict	2.30	0.179	0.011	0.098	0.116	0.190	**0.312**	**0.411**	0.534	0.774	**0.584**	**0.728**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Reynolds (1D Scalar)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Reynolds (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

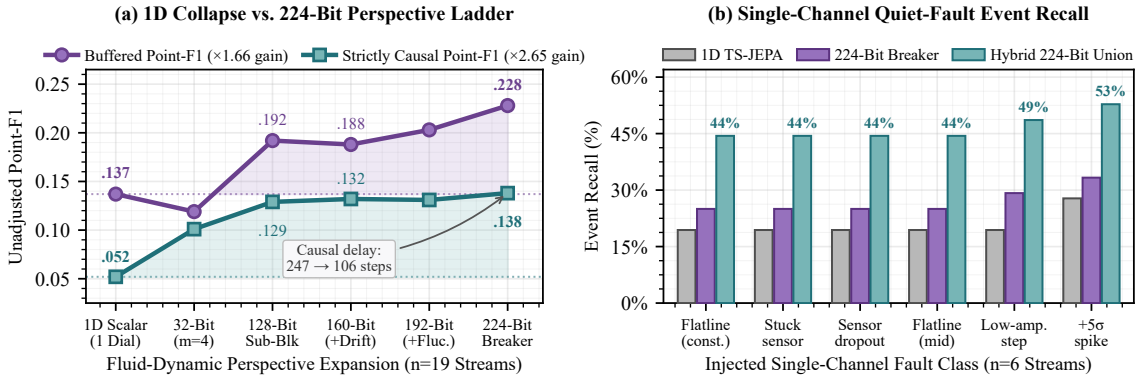


Fig. 3. Perspective expansion and quiet-fault recovery. **(a)** Causal and buffered Point-F1 across the 6-step fluid-dynamic expansion from 1D scalar collapse to the 224-bit circuit-breaker panel (n=19 streams). **(b)** Single-channel fault event recall comparing 1D TS-JEPA, standalone 224-Bit Breaker, and the Hybrid 224-Bit Union (n=6 streams).

`SMD machine-1-2`, `Daphnet S01R01E1`, and `GECCO water-quality`, three seeds). The dyadic front-end helps on high-channel spacecraft telemetry (buffered Point-F1 0.085 → 0.410 on `MSL M-1`, $K = 55$; causal 0.212 → 0.321 on `SMAP P-3`, $K = 25$) but hurts on $K = 9$ streams (0.136 vs. 0.217 on `Daphnet`, 0.158 vs. 0.321 on `GECCO`), so the plain stem wins on 5-stream macro causal Point-F1 (0.193 vs. 0.156). Shell splitting pays off only on high-channel telemetry.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 15 model arms across the 19 non-GHL streams under identical splits, training budgets, and matched parameter counts in Table II(c) isolates five architectural factors:

1) *Forecasting vs. reconstructing raw observations (TCN Obs. Forecaster vs. TCN Obs. Autoencoder):* Pairing the same causal dilated TCN encoder with a multilayer perceptron decoder to either forecast the 64-step target window or reconstruct the 320-step window in raw sensor space leaves tail kurtosis ($\gamma_2 \approx 6.2$), calibration drift ($D_{KS} \approx 0.45$), and causal FPR ($\approx 25\%$) nearly identical.
2) *LayerNorm bottleneck in observation space (TCN Obs. Forecaster (LN)):* Adding a $D = 32$ LayerNorm bottleneck and target-window LayerNorm to the observation forecaster compresses tail kurtosis six-fold ($\gamma_2 = 0.91$) and halves causal FPR to 11.7%, but targeting raw sensor waveforms leaves calibration drift high ($D_{KS} = 0.376$) and drops causal Point-F1 to 0.084.
3) *Learned latent target projection, multi-horizon whitening, and 224-bit circuit breakers (TS-JEPA and Multi-Horizon Reynolds 224-Bit):* Moving the prediction target into the learned $D = 32$ latent space cuts calibration drift nearly in half ($D_{KS} = 0.214$ in TS-JEPA and 0.179 in Multi-Horizon Reynolds JEPA, paired Wilcoxon $p \leq 0.001$).
4) *Nonlinear TCN vs. linear PCA and Isolation Forest:* 32 principal components with ridge regression keep causal FPR at 2.1% but causal Point-F1 falls to 0.017, and

Isolation Forest reaches only 0.149 causal event recall.

5) *Prediction vs. context autoencoding, frozen targets, and ℓ_2 scoring:* Latent context autoencoding keeps causal FPR at 1.7% but lowers causal Point-F1 from 0.161 to 0.112; a frozen target encoder or ℓ_2 scoring still holds causal FPR below 2%, so the false-alarm reduction comes from the normalized latent representation.

VII. Results and Discussion

A. Primary 33-Stream and Archive-Wide 141-Stream Benchmark Results

Table III evaluates all 33 primary continuous telemetry streams under the two-stage chronological held-out protocol: Panel (a) reports retrospective buffered attribution, Panel (b) strictly causal streaming, and Panel (c) per-dataset breakdowns including anomaly event counts (E) and held-out nominal FPRs. Archive-wide 141-stream results appear in Table V; on a 130-stream baseline screen, TS-JEPA achieves 0.186 stream and 0.160 macro Point-F1 versus 0.152 for TranAD, 0.151 for TimesNet, 0.097 for DCdetector, 0.088 for NCAD, and 0.082 for Anomaly Transformer. Across both attribution regimes, pairing Multi-Horizon Reynolds JEPA with the 224-bit circuit-breaker panel achieves the highest buffered Point-F1 (0.228), buffered PR-AUC (0.411), causal PR-AUC (0.312), causal event recall (0.607), and Dataset-Macro Point-F1 (0.122), while keeping held-out false alarms an order of magnitude below observation-space reconstructors (1.1%–1.7% causal and 9.8%–13.2% buffered vs. 22.2%–30.7% for TimesNet and TranAD). Its causal Point-F1 (0.138) is below TimesNet (0.209) and TS-JEPA (0.161), so on these streams its advantage is a better balance of false alarms and recall, not a uniform win.

Reducing Observation-Space False-Alarm Flooding: On the 19 non-GHL primary streams (Table III(a),(b)), predicting spherical latent representations cuts held-out nominal FPR three-fold under buffered attribution to 7.8%–9.8% versus 30.0%–30.7% for TimesNet and TranAD ($p \leq 0.045$), and by an order of magnitude under strictly causal streaming to 0.6%–1.7% versus 22.2%–24.3% for reconstructors ($p = 0.010$). This separation holds across all $N = 7$ dataset families ($p = 0.016$) and under zero-tuning LODO evaluation, where JEPAs maintain 0.3%–2.3% causal FPR versus 12.1%–18.4% for reconstructors.

Replacing the Scalar Score: Under 1D scoring, Multi-Horizon Reynolds JEPA fires very conservatively (0.050 causal and 0.148 buffered Point-F1) because one or two drifting coordinates inflate the scalar threshold. On the same trained networks, the panel more than doubles causal Point-F1 to 0.116–0.138 ($17/0$ wins, $p = 1.5\times10^{-5}$) and cuts causal delay from 246.6 to 105.9 steps (Section VI). On GHL ($\pi = 0.013$), all JEPAs and the linear autoregressive (AR) forecaster fire near-silently (0.1%–0.4% FPR, 0.000 F1): low FPR without recall is not detection.

Scope of Applicability: The evidence supports the panel only under four conditions: (i) multichannel arrays ($K \geq 9$; SMAP, MSL, Daphnet, SMD, Exathlon, OPPORTUNITY); (ii) sampling rates at which a 63-step emission lag is acceptable, because the panel was evaluated only at $S = 64$ (Section VII-C), which rules out hourly Metro (63 steps $= 63$ h); (iii) calibration blocks of roughly 10^3 steps or more (Section VII-E); and (iv) faults that move the latent summary. It fails on the slow process ramps of GHL and Metro (0.000–0.022 F1) and on single-channel freezes absorbed by LayerNorm, where the panel alone catches 25.0%–29.2% of events and a rolling variance monitor is needed to reach 44.4%–52.8% (Section VI-B).

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Comparing Panels (a) and (b) of Table III shows why the panel matters most for real-time streaming. Without 64-step averaging, causal whitened residuals have a heavier right tail than buffered scores ($\gamma_2 = 2.30$ vs. 1.25). Under 1D scoring, hijacker dials push the threshold onto that tail, so 1D Multi-Horizon Reynolds JEPA alerts on only 2.4% of test steps and flags isolated peaks inside each anomaly (0.050–0.052 causal F1). With per-dial switches and onset relays (7), *224-Bit Breaker* stays active across the fault window (0.138 causal F1, 60.7% event recall, 105.9-step delay, 1.7% FPR). A rate-matched random alerter, $\mathrm{ER}_{\mathrm{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$, would catch 28.1% of events at the panel's nominal rate, but 60.0% at TimesNet's 24.3% FPR, above TimesNet's own 58.8%.

C. Horizon Length, Structural Emission Lag, and Fast-Tripwire Regimes

Target horizon $S = 64$ imposes a structural emission lag of $S - 1 = 63$ steps before a causal trailing window decision is emitted at $\tau = t + S - 1$ (Section IV-D). To evaluate detection quality versus latency against baselines at matched emission lag $S - 1 \in \{0, 3, 15, 31, 63\}$, Fig. 4(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams at three seeds. At $S = 1$ (zero lag) and $S = 4$ (3-step lag), latent prediction keeps causal FPR at 0.4%–0.6% versus 2.5%–3.1% for TimesNet, with higher causal PR-AUC (0.178–0.189 vs. 0.150–0.162) and short delays ($\Delta t = 57.8$–85.2 steps). As S increases from 16 to 64, TimesNet causal FPR climbs from 6.1% to 21.3%, whereas latent predictors remain bounded at $\leq 2.0\%$. Thus, $S = 64$ suits high-rate telemetry (≥ 1 Hz, such as 64 Hz Daphnet where 63 steps $= 0.98$ s) or subsystem trend monitoring on 0.1 Hz SMAP and MSL (10.5 min), while low-rate streams ($\leq 1\,\mathrm{min}^{-1}$) and zero-lag tripwires require $S \in \{1, 4\}$. These short-horizon runs use 1D latent scoring: the panel needs four nested prefixes and has not been evaluated below $S = 64$.

D. Operating Budgets and Contextual vs. Point Anomalies

Because reconstruction baselines calibrate to 22%–31% realized FPR versus 0.6%–2.4% for latent predictors, we evaluate three operating-point comparisons:

1) *Autonomous circuit-breaker points and nominal calibration quantiles (q^*, Fig. 4(b), Table II(c)):* Without test labels, Multi-Horizon Reynolds 224-Bit Strict (53.4% ER at 1.1% FPR) and 224-Bit Breaker (60.7% ER at 1.7% FPR) dominate the low-FPR frontier; at calibration quantile

TABLE III
HELD-OUT MULTI-STREAM TELEMETRY BENCHMARK ACROSS ATTRIBUTION REGIMES AND DATASETS (BEST IN **BOLD**, SECOND UNDERLINED).

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-perspective fluid-dynamic circuit-breaker architectures												
Reynolds 224-Bit Breaker	0.132	**0.228**	**0.411**	**0.631**	71.8	0.017	0.138	**0.607**	105.9	**0.002**	0.000	**0.122**
Reynolds 224-Bit Strict	0.098	0.190	**0.411**	0.545	97.3	0.011	0.116	0.534	248.0	**0.002**	0.000	0.098
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Reynolds (1D scalar)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Reynolds (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	$F1_{all}$	**224-Bit**	**TS-JEPA**	**Rey. 1D**	**TimesNet**	**TranAD**	**224-Bit**	**TS-JEPA**	**Rey. 1D**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.223	0.145	0.143	**0.229**	0.144	**.007/.047**	.030/.091	.008/.072	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.001/.002**	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	**0.063**	0.000	0.012	0.000	0.000	.010/.038	.004/.017	.003/.098	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.199**	0.161	0.081	0.158	0.151	.014/.131	.054/**.095**	**.004**/.098	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.354	**0.379**	0.267	0.289	0.167	.021/.214	.014/.135	**.006/.093**	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	**0.001**	**0.001**	0.000	.002/.007	.008/.014	.002/.007	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.017	**0.119**	0.008	0.000	0.000	**.000/.000**	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**

Note: In (c), n = streams, K = channels, E = events, π = anomaly rate, $F1_{all} = 2\pi/(1+\pi)$. 95% stream-bootstrap CIs (n=19): 224-Bit Breaker buffered FPR [.068, .198], F1 [.142, .321] and causal FPR [.008, .027], F1 [.068, .224] (vs. Reynolds 1D causal F1 [.013, .097] and TimesNet causal FPR [.123, .375]).

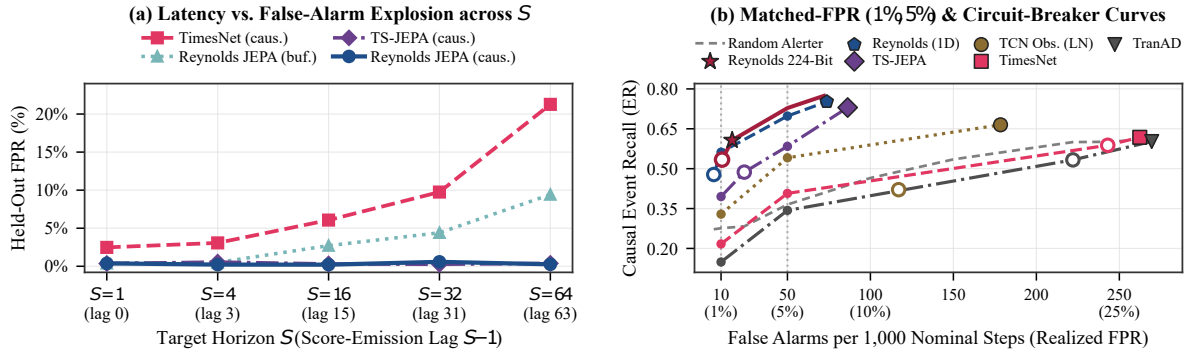


Fig. 4. Horizon length and operating trade-offs. **(a)** Held-out FPR across target horizons $S \in \{1, 4, 16, 32, 64\}$ at matched emission lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps (n=5 streams). **(b)** Causal event recall vs. false alarms across n=19 streams at matched test-FPR budgets $f^* \in \{1\%, 5\%\}$ (vertical dotted guides), autonomous 224-Bit Strict (open circle) and 224-Bit Breaker (filled star) operating points, and calibration quantiles $q^* \leq 0.05$.

TABLE IV
CONTEXTUAL VS. POINT ANOMALY DETECTION (EVENT RECALL (ER)).

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**Auto. ER**	**FPR**	f^*=.01	f^*=.05	f^*=.01	f^*=.05
Reynolds 224-Bit	**0.658**	**0.750**	0.018	**0.820**	**0.920**	0.683	0.944
Reynolds (1D)	0.652	**0.750**	**0.006**	**0.820**	**0.920**	0.683	0.867
Reynolds (α_{LT}=0)	0.641	0.625	**0.006**	**0.820**	**0.920**	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=0.05	f^*=0.01	f^*=0.05	q^*=0.05	f^*=0.01	f^*=0.05
Reynolds 224-Bit	**0.654**	**0.552**	**0.618**	**0.796**	0.688	**0.792**
Reynolds (1D)	0.628	0.498	0.590	0.782	0.683	0.781
Reynolds (α_{LT}=0)	0.640	0.540	0.603	0.785	0.671	0.785
TS-JEPA	0.614	0.520	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Panel (a) stratifies the 10 primary SMAP/MSL streams by NASA's ground-truth taxonomy [4] (contextual: SMAP E-1, A-2, A-7, MSL M-1, and C-1 [2100, 2210]; point: SMAP A-1, P-3, D-3, MSL D-14, D-15, and C-1 [550, 750]). Panel (b) stratifies all n=19 non-GHL streams into in-range contextual ($\max|z| \leq 5\sigma$) vs. out-of-range point ($\max|z| > 5\sigma$) causal events.

$q^* = 0.050$, Multi-Horizon Reynolds 224-Bit reaches 77.4% causal ER versus 61.8% for TimesNet and 60.2% for TranAD.

2) *Post-hoc matched realized test-FPR budgets (f^*, Fig. 4(b), Table II(c)):* This diagnostic sets each threshold from test labels, so no deployed system could reach these points; it only checks that the low false-alarm rates are not an artifact of low alert frequency (vertical dotted guides in Fig. 4(b)). At a 1% test FPR budget ($f^* = 0.01$), Multi-Horizon Reynolds 224-Bit detects 58.4% of anomaly events (56.2%–57.8% for 1D Multi-Horizon Reynolds and 39.5% for TS-JEPA), nearly tripling TimesNet (21.7%) and quadrupling TranAD (14.9%), with a similar gap at $f^* = 0.05$ (72.8% vs. 34.3%–40.7%).

3) *Contextual vs. point anomaly stratification (Table IV):* Point anomalies ($\pm 5\sigma$ spikes) are caught even by a marginal z-score (`ZScoreMax`); contextual anomalies stay in range and break multi-step structure. On the 10 primary SMAP/MSL streams stratified by NASA's taxonomy [4] (Table IV(a)), observation-space detectors collapse on contextual anomalies (0.161–0.317 buffered PR-AUC). Latent JEPAs lead on these five contextual sequences; the 224-bit

TABLE V
ARCHIVE-WIDE CIRCUIT-BREAKER BENCHMARK (3 SEEDS): 118 HELD-OUT STREAMS NEVER USED IN DEVELOPMENT, FULL 141-STREAM AGGREGATE, AND PER-DATASET BREAKDOWN.

	Pt-F1 ↑		PR-AUC ↑		ER ↑		FPR_{hold} ↓	
(a) Held-Out $n{=}118$	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
TS-JEPA (1D)	.072	.142	.154	.202	.484	.458	**.051**	**.104**
Speculative-JEPA (1D)	.076	.148	.172	.245	.509	.577	.054	.194
Spec.-Alien-JEPA (1D)	.083	.146	.174	.242	.512	.573	.059	.199
Spec.-224Bit-Strict	.134	.166	**.183**	**.247**	.662	.673	.073	.181
Spec.-224Bit-Breaker	**.143**†	**.168**‡	**.183**	**.247**	**.728**	**.691**	.083	.210
All $n{=}141$ **(incl. 23 dev.)**								
Spec.-Alien-JEPA (1D)	.077	.153	.184	.253	.507	.567	.051	.180
Spec.-224Bit-Breaker	.139	.180	.197	.261	.727	.685	.072	.196

		Causal Pt-F1 ↑			Buffered Pt-F1 ↑			Causal FPR_{hold} ↓		
(b) Held-out	n	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**
SMAP	40	**.194**	.129	.091	.201	.193	**.214**	.088	.086	**.065**
SMD	18	**.090**	.028	.042	**.125**	.065	.072	.019	**.006**	.009
Exathlon	17	**.163**	.133	.092	.181	**.186**	.134	.235	.137	**.119**
Daphnet	15	**.125**	.053	.075	**.153**	.125	.105	.085	**.063**	**.063**
MSL	13	**.133**	.051	.074	**.239**	.176	.191	.023	**.007**	.011
OPPORT.	12	**.075**	.018	.024	.066	**.067**	.030	.027	**.003**	.008

†vs. 1D Spec.-Alien: $\Delta{=}+0.060$ (95% CI $[+0.042, +0.079]$), $95/15$ wins, $p{=}1.6{\times}10^{-13}$, positive in 8 of 9 datasets (sign-flip $p{=}0.008$); vs. TS-JEPA $85/28$, $p{=}6.4{\times}10^{-11}$. ‡vs. 1D Spec.-Alien: $\Delta{=}+0.021$ ($[+0.010, +0.034]$), $p{=}1.7{\times}10^{-4}$, dataset-level $p{=}0.27$. In (b), 224B = Spec.-224Bit-Breaker, 1D-Al. = 1D Spec.-Alien, TS = TS-JEPA (115 of 118 streams; three single-stream datasets in text).

panel matches 1D Multi-Horizon Reynolds JEPA (0.658 vs. 0.652 buffered PR-AUC, 75.0% autonomous recall, and identical 82.0%–92.0% recall at $f^* \in \{0.01, 0.05\}$) at a higher autonomous FPR (1.8% vs. 0.6%), so this evidence is preliminary. Across all 19 non-GHL streams in Table IV(b), Multi-Horizon Reynolds 224-Bit doubles reconstructor recall on contextual faults (61.8% vs. 30.0% at $f^* = 0.05$) and leads on point faults (79.2% vs. 56.3%–61.1%).

E. Archive-Wide Held-Out Analysis, Calibration Length, and Drift Decomposition

Table V evaluates the panel on the three-horizon Speculative-JEPA (192 distinct dials, Section IV-B) across 141 streams in 12 datasets at three seeds; 118 of these streams were never used to tune the panel:

1) *Held-out generalization on 118 streams (Table V(a)):* The panel constants were frozen before these streams were scored, so they are the primary test. Under 1D scoring, moving from TS-JEPA to three-horizon Speculative-JEPA raises PR-AUC (0.154 → 0.172 causal, 0.202 → 0.245 buffered). Replacing the scalar score with the panel then raises causal Point-F1 from 0.072–0.083 to 0.134–0.143 and causal event recall from 48.4%–51.2% to 66.2%–72.8%, and cuts causal delay from 84.7–107.8 to 55.8–77.0 steps. The cost is more false alarms: causal held-out FPR rises from 5.1%–5.9% to 7.3%–8.3%. Under buffered attribution the Point-F1 gain shrinks to +0.021 and is not significant across datasets. The 141-stream aggregate, which includes the 23 development streams, shows slightly larger gains (causal +0.062, dataset-cluster 95% CI $[+0.044, +0.072]$, sign-flip $p = 0.007$, at least +0.057 under every leave-one-dataset-out split).
2) *Per-dataset consistency (Table V(b)):* The panel leads causal Point-F1 in all six multi-stream held-out datasets, most on SMAP (0.194 vs. ≤ 0.129) and SMD (0.090 vs. ≤ 0.042), but it also has the highest causal FPR in each, most on Exathlon (23.5% vs. ≤ 13.7%). Under buffered attribution it trails the best 1D JEPA on SMAP, Exathlon, and OPPORTUNITY. On CreditCard, GECCO, and PSM (one stream each) all methods lie within 0.007 causal Point-F1.
3) *Calibration length:* Every quantile in (6)–(8) is estimated from overlapping, autocorrelated windows. On the 19 primary streams the effective sample size of the calibration dials (integrated autocorrelation) is about 0.31 of the window count (median 152 of 408), so $q_{0.999}$ rests on few independent observations. On the 118 held-out streams, mean causal FPR is 19.2% for the 12 streams with fewer than 250 calibration steps and 1.6% for the 22 with at least 2,500. Cutting the nominal block of five long primary streams into short sub-streams confirms the trend: causal held-out FPR is 18.9%, 8.3%, 2.6%, and 4.1% at 100, 250, 500, and 1,000 calibration windows versus 0.5% on the full block, and the coefficient of variation of K_{224} falls from 0.34 to 0.10. 1D SPOT on the same backbone stays at or below 1.2% from 250 windows on. The many per-dial quantiles therefore need roughly 10^3 calibration steps; at the $N_{\text{cal}} \geq 100$ inclusion floor the panel's FPR is not reliable.
4) *Three-stage permutation drift decomposition:* Refitting thresholds across the 19 primary streams under chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and pointwise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) splits confirms that inter-block regime drift ($\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$) drives 71.3% of excess causal FPR in TS-JEPA (0.0229 vs. 0.0073) and 78.3%–81.8% in reconstructors (0.210–0.230 vs. 0.043–0.046), whereas barycenter centering in Multi-Horizon Reynolds JEPA cuts the inter-block drift share four-fold to 18.7% (0.0056 vs. 0.0047) while per-dial hijack guards γ_j quarantine individual drifting coordinates.

VIII. CONCLUSION AND FUTURE WORK

We studied whether Joint-Embedding Predictive Architectures and multi-perspective circuit-breaker panels reduce observation-space false-alarm flooding and 1D scalar score collapse across 33 primary and 141 archive-wide telemetry streams:

1) *A breaker panel in place of a scalar threshold:* Expanding $D = 32$ whitened coordinates into 224 per-coordinate horizon statistics with nominal drift guards, prefix-routed attribution, and 32 onset relays keeps one or two drifting coordinates from setting a shared threshold. On $n{=}19$ primary streams, causal Point-F1 rises from 0.052 to 0.138 ($17/0$ wins, $p = 1.5{\times}10^{-5}$) and causal delay falls from 246.6 to 105.9 steps; on the 118 held-out archive streams, causal Point-F1 rises from 0.072–0.083 to 0.143 ($95/15$ wins, $p = 1.6{\times}10^{-13}$) at 72.8% recall, while causal FPR rises from 5.9% to 8.3%.
2) *Factorial Mechanism Isolation and Component Verification:* Matched-encoder ablations isolate LayerNorm kurtosis compression (γ_2 : 6.10 → 0.91), learned latent projections (D_{KS} : 0.376 → 0.214), and barycenter centering in Multi-Horizon Reynolds JEPA (RankMe : 11.6 → 19.8, $D_{\text{KS}} = 0.179$, inter-block drift share 71.3% → 18.7%). Leave-one-out ablations support the right-edge relays (−17.8-step causal delay) and prefix

routing ($+4.7\%$ buffered recall); hijack guards add little on these streams ($0.137 \rightarrow 0.138$ causal F1), scalar spectral-concentration modulation shows no measurable benefit, and short horizons ($S \in \{1, 4\}$) remove the 63-step emission lag.

3) *Limits:* At a 1% test-FPR budget the panel catches 58.4% of events versus 21.7% for TimesNet, but gains concentrate on multichannel streams. It misses the slow process ramps of GHL and Metro (0.000–0.022 Point-F1), catches only 25.0%–29.2% of single-channel freezes alone (44.4%–52.8% with a variance monitor, on six injected streams), and its contextual-anomaly evidence rests on five NASA sequences. The panel needs about 10^3 calibration steps to hold its FPR. The archive benchmark interpolated the 48-step horizon and reused its calibration block for early stopping; a four-horizon rerun with a disjoint validation block, calibration-constant sensitivity, and online adaptation for non-stationary ramps remain future work.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.

[4] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[8] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[10] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[11] O. Reynolds, "On the dynamical theory of incompressible viscous fluids and the determination of the criterion," *Philosophical Transactions of the Royal Society of London A*, vol. 186, pp. 123–164, 1895.

[12] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.

[13] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[14] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.

[15] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.

[16] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[17] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[18] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[19] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[20] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[21] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[22] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[23] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[24] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[25] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[26] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[27] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[28] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[29] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[30] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[31] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[32] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[33] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[34] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[35] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.

[36] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[37] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[38] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[39] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[40] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[41] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[42] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[43] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[44] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[45] Q. Garrido, R. Balestriero, L. Najman, and Y. LeCun, "RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank," in *International Conference on Machine Learning (ICML)*, 2023, pp. 10 929–10 974.

[46] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[47] E. H. Lieb and W. E. Thirring, "Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities," in *Studies in Mathematical Physics: Essays in Honor of Valentine Bargmann*. Princeton University Press, 1976, pp. 269–303.

[48] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.

[49] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[50] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.